Bulk Reconstruction: The Interior Two-Point Function from the Cone Data

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Abstract. The assembly paper identified the state built from this series' ingredients with the light-cone current data of the 3+1 Minkowski massless vacuum, and named the cone's interior as the next phase. This note reconstructs the interior, at the two-point level, completely — with proofs, not citations. Theorem 1 (the rest-frame reconstruction formula): for an interior point on the apex axis, $x = (t, \vec{0})$, every generator meets x's past cone at $\lambda = t/2$, and $\phi(x) = \frac{1}{4\pi} \oint d\Omega_{\hat{n}}\, (\partial_\lambda \psi)(t/2, \hat{n})$ — proved by a plane-wave computation that collapses to a total derivative, verified numerically at 10^{-11} across random null momenta. Theorem 2 (the mixed-correlator identity): the same formula, applied in one argument against an *arbitrary* interior second point, reproduces the bulk Wightman function exactly — the sphere integral evaluates in closed form, $\oint d\Omega\, (A + Bc)^{-2} = 4\pi/(A^2 - B^2)$, and the algebra closes identically, $A^2 - B^2 = y^2 \left[(T - t_1)^2 - |\vec{y}|^2\right]$; verified at 10^{-13} across random interior points. Corollary: since every interior point lies on a timelike line through the apex — its own — and the assembled state is Lorentz invariant, each argument reconstructs in its own apex frame: **the full interior two-point function, for all pairs of interior points, is determined by and computed from the assembled cone data**, constants included — the chain from the generator-diagonal chiral density $-\delta^2(\hat{n}, \hat{m})/4\pi(\lambda - \mu - i\epsilon)^2$ to the interior $-1/4\pi^2\tau^2$ closes at 0.0. Higher correlations follow from quasi-freeness on both sides. The interior is not new data: it is the cone data read through a geometric formula.

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1. The task and the inputs

The assembly paper established: the assembled state — axis vacua on the complete null geodesics through the apex, generator-diagonal, with per-generator kernel $-1/4\pi(\lambda - \mu - i\epsilon)^2$ — is the light-cone current data of the Minkowski massless vacuum, and it is exactly Lorentz invariant. What remained was the interior: do the cone data determine correlations at points *inside* the cone, and can the determination be exhibited rather than asserted? Both questions are answered affirmatively here, by two theorems whose proofs are elementary enough to give in full.

Throughout, $\psi = \lambda\phi$ is the null-projected field of the assembly paper, $\partial_\lambda \psi$ its current data, and for an interior point x and direction $\hat{n}$, the generator $\lambda \mapsto \lambda(1, \hat{n})$ meets x's past light cone at the single value $\lambda_*(x, \hat{n}) = x^2/2(x^0 - \vec{x} \cdot \hat{n})$.

2. Theorem 1: the rest-frame reconstruction formula

Theorem 1. For $x = (t, \vec{0})$ inside the cone, $\lambda_*(x, \hat{n}) = t/2$ for every $\hat{n}$, and

$$\phi(x) \;=\; \frac{1}{4\pi} \oint_{S^2} d\Omega_{\hat{n}}\; (\partial_\lambda \psi)\big(\tfrac{t}{2}, \hat{n}\big).$$

Proof. By linearity it suffices to check plane waves $\phi_k(x) = e^{-ik\cdot x}$, $k = \omega(1, \hat{k})$ null. On the cone, $\psi_k(\lambda, \hat{n}) = \lambda\, e^{-i\omega\lambda(1-c)}$ with $c = \hat{n} \cdot \hat{k}$, so with $s = 1 - c \in [0, 2]$,

$$\oint d\Omega\, (\partial_\lambda \psi_k)\big(\tfrac{t}{2}, \hat{n}\big) = 2\pi \int_0^2 \left[1 - i\omega\tfrac{t}{2}s\right] e^{-i\omega\frac{t}{2}s}\, ds = 2\pi \int_0^2 \partial_s\!\left[s\, e^{-i\omega\frac{t}{2}s}\right] ds = 4\pi\, e^{-i\omega t} = 4\pi\, \phi_k(x$$

Verified numerically for random (ω, t): deviations $4.6 \times 10^{-12}, 8.3 \times 10^{-12}, 3.6 \times 10^{-11}$ against targets of modulus one. The geometric content: the past cone of an apex-axis point cuts the cone in the round sphere $\lambda = t/2$, and the field at x is the *unweighted average* of the current data over that cut.

3. Theorem 2: the mixed-correlator identity, general second point

Theorem 2. Let $x = (t_1, \vec{0})$ and let $y = (T, \vec{y})$ be any interior point with the cut sphere of x timelike-separated from y. Then

$$\frac{1}{4\pi} \oint d\Omega_{\hat{n}}\; \partial_\lambda \big[\lambda\, W(\lambda(1,\hat{n}), y)\big]_{\lambda = t_1/2} \;=\; W(x,y).$$

Proof. On a generator, $(\lambda(1,\hat{n}) - y)^2$ is *linear* in λ (the λ^2 terms cancel on null directions): $W(\lambda(1,\hat{n}), y) = -1/4\pi^2 \left[y^2 - 2\lambda D(\hat{n})\right]$ with $D(\hat{n}) = T - \vec{y}\cdot\hat{n}, y^2 = T^2 - |\vec{y}|^2$. Hence $\partial_\lambda[\lambda W] = -y^2/4\pi^2 \left(y^2 - 2\lambda D\right)^2$. At $\lambda = t_1/2$, with $A = y^2 - t_1 T$, $B = t_1|\vec{y}|$, the sphere integral is elementary: $\oint d\Omega\, (A + Bc)^{-2} = 4\pi/(A^2 - B^2)$. And the identity

$$A^2 - B^2 = (y^2 - t_1 T)^2 - t_1^2|\vec{y}|^2 = y^4 - 2t_1 T y^2 + t_1^2 y^2 = y^2\left[(T - t_1)^2 - |\vec{y}|^2\right]$$

closes the computation: the left side equals $-y^2/4\pi^2(A^2 - B^2) = -1/4\pi^2\left[(T - t_1)^2 - |\vec{y}|^2\right] = W(x,y)$. ■

Verified numerically for three random interior y: relative deviations $2.3 \times 10^{-13}, 5.0 \times 10^{-13}, 2.9 \times 10^{-13}$.

4. Corollary: the full interior two-point function

Corollary. The two-point function at *every* pair of interior points is determined by, and computed from, the assembled cone data.

Argument. Two steps, one boost each — never both bulk points on one axis. **Step 1.** Every interior point lies on a timelike line through the apex — its own. Apply the single boost taking x to the axis, $x = (t_1, \vec{0})$; the second point goes to an arbitrary interior y, and correlator values are unchanged by the established invariance. Theorem 1 is an identity of solutions, so it holds slot-by-slot inside correlation functions of the free field: the first leg reduces,

$$W(x,y) = \frac{1}{4\pi} \oint d\Omega_{\hat{n}}\, \big\langle (\partial_\lambda \psi)(t_1/2, \hat{n})\, \phi(y) \big\rangle,$$

and Theorem 2 is precisely the certification that this reduction reproduces the correct value against arbitrary y. **Step 2.** The remaining bulk leg is reduced by Theorem 1 applied in y 's own apex frame. This is licensed transport, not a frame change mid-formula: the cone and its data form a Lorentz-covariant family, so under the boost adapted to y the first leg *remains a cone observable* — the cut sphere of x is a frame-independent subset of the cone, its smeared current merely relabeled — while the second slot becomes an axis point to which Theorem 1 applies. The result expresses $W(x,y)$ as a double sphere integral of pure cone-data correlations, each leg reduced in the frame adapted to it, the two frames linked only through the covariant cone family. (What is *not* claimed is a single-frame formula with the uniform measure on both legs at once; the covariant weight of such a general-frame formula is derivable from the boost's conformal action on the sphere and is not needed here.) The constants chain from the assembled data closes exactly: inserting the generator-diagonal density $-\delta^2(\hat{n}, \hat{m})/4\pi(\lambda - \mu - i\epsilon)^2$ into the double reconstruction integral for two apex-axis points — the one-frame, collinear case — gives

$$\frac{1}{16\pi^2} \cdot 4\pi \cdot \left(-\frac{1}{4\pi(\Delta t/2)^2} \right) = -\frac{1}{4\pi^2\,\Delta t^2},$$

the interior Wightman function at timelike separation Δt — verified at 0.0. Higher correlation functions follow on both sides from quasi-freeness. ■

5. What this establishes for the series

The bridge now reaches the interior. From the Fano plane through the contraction tower, the unique squeeze-thermal state, the type III_1 verdict, the local net, the exponential correspondence, and the light-cone identification, the series arrives at: interior correlations of a 3+1 massless field, derived — not matched, derived, through proved formulas with the constants closing at machine precision — from data the model assembled out of its own ingredients. And the *form* of the result keeps faith with the series' relational stance: the interior field is not new substance. $\phi(x)$ is the unweighted average of current data over the sphere in which x's past cone cuts the assembly — the interior point *is* a relation among cone observables, exactly as positions were relations among operators and time a relation borne by the state.

6. Scope

Five items. *(i)* Reconstruction is established at the two-point level with quasi-free closure stated, not re-verified; this determines the free theory. *(ii)* The operator-algebraic refinement — interior local *algebras* generated from cone algebras, isotony and locality of the interior net — is the named next step; the present result is the correlation-function core it will rest on. *(iii)* The identities are verified at timelike-regular configurations; distributional boundary values carry the $-i\epsilon$ prescription of the assembly paper throughout. *(iv)* The soft, cross-generator sector remains excluded: reconstruction here concerns the hard sector, and charges and memory still live at the frontier named previously. *(v)* Everything is free and massless; interactions remain beyond, with the series' standing audits (Lorentz rerun, curvature as entropic response) waiting there.

References

- Companion papers: *The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output*; *Geometrization: The Exponential Correspondence, Thermal Relative Duality, and the Shadow's Home*; *Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group*; *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III_1 Verdict*.
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- Sewell, G. L., "Quantum fields on manifolds: PCT and gravitationally induced thermal states," *Ann. Phys.* 141 (1982), 201–224.
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